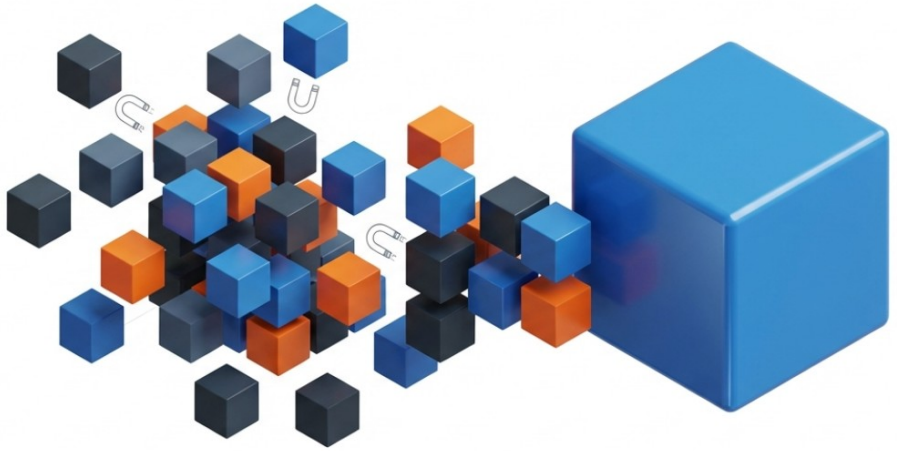
CHAPTER 4

C++ PROGRAMMING



Trường Đại học Bách Khoa
Khoa/Viện: [Tên đơn vị]
Giảng viên: [Họ tên]

Dùng cho bài giảng, seminar, bài tập lớn hoặc thuyết trình học thuật.

01

REPETITION STRUCTURES IN C++

Using loops to execute statements repeatedly based on a condition

Learning Objectives

1. **Understand repetition structures**
   Distinguishing pre-test and post-test loops, fixed-count loops, and condition-controlled loops

2. **Write basic C++ loops**
   Use while, for and do-while statements correctly

3. **Apply loops to programming problem**
   Use counters, sentinel values, accumulators, and nested loops

4. **Control loop execution**
   Use break and continue, and identify common causes of infinite loops

LESSON ROADMAP

From basic concepts -> syntax -> examples -> practice

1

Loop concepts

Pre-test and post-test
Fixed-count and condition-controlled

2

Basic Loop Statements

While, for, do-while

3

Advanced Repetition

Sentinel Values, Nested Loops

4

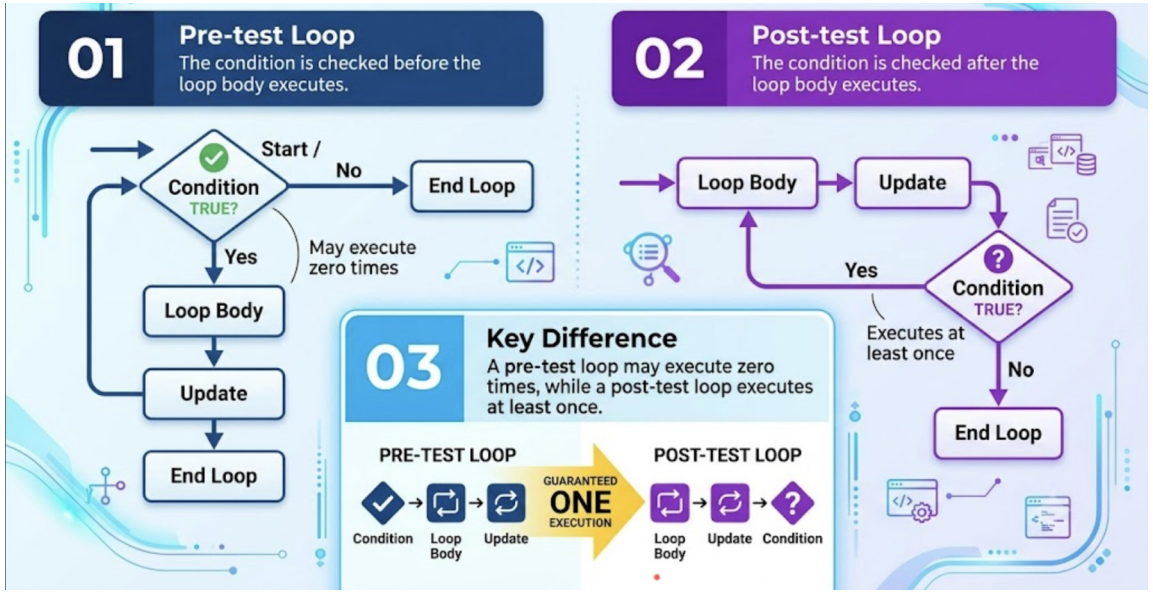
Loop Control & Practice

Break, continue, infinite loops and exercises

What is a Repetition Structure?

A Repetition Structure executes a statement or a group of statements repeatedly while a specified condition allows execution to continue

C++ provides three main repetition structures: while, for and do-while.
The condition can be tested either before or after the loop body.



The position of the condition check determines whether the loop body is guaranteed to execute

01 **Pre-test Loop**
The condition is checked before the loop body executes

02 **Ý chính 2**
Mô tả ngắn / ví dụ

The While Statement

Concept

- The while statement repeats a statement or a group of statements as long as the condition evaluates to true.
- Syntax:

```
while (condition) {
    statements;
}
```

- **Three important components**
  Initialize the control variable.
  Check the loop condition.
  Update the control variable.
- while is a **pre-test loop**. If the condition is false initially, the body is not executed.

Example

- Print integers from 1 to 10

```
int count = 1;

while (count <= 10) {
    cout << count << " ";
    count++;
}
```

Output

```
1 2 3 4 5 6 7 8 9 10
```

- What happens if count++ is removed?
  The loop may become infinite.

Fixed-Count vs. Condition-Controlled Loops

FIXED-COUNT LOOP

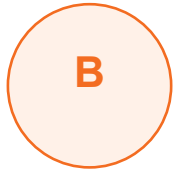


Main Characteristics

- The number of iterations is known in advance.
- Usually controlled by a counter.
- The counter changes after each iteration.
- for is commonly suitable.

```
for (int i = 1; i <= 10; i++)
```

CONDITION-CONTROLLED LOOP



Main Features

- The number of iterations is not known in advance.
- Execution depends on changing input or state.
- Often uses a sentinel value.
- while is commonly suitable.

```
while (grade <= 100)
```

Use a fixed-count loop when the number of repetitions is predictable.
Use a condition-controlled loop when termination depends on runtime data.

Sentinel-Controlled Repetition

INPUT GRADE	VALID?	ACTION
75	YES	ACCEPT
90	YES	ACCEPT
68	YES	ACCEPT
105	NO	STOP

```
cin >> grade;

while (grade <= 100) {
    cout << "Grade accepted\n";
    cin >> grade;
}
```

How a for Loop Executes

```cpp
for (int count = 2; count <= 20; count += 2) {
    cout << count << " ";
}
```



What if we never update the "count" – never count +=2; ?

NESTED LOOPS

```
for (int i = 1; i <= 5; i++) {
    for (int j = 1; j <= 4; j++) {
        cout << "i=" << i
             << ", j=" << j << endl;
    }
}
```

```
Outer loop: i = 1
   ├── j = 1
   ├── j = 2
   ├── j = 3
   └── j = 4

Outer loop: i = 2
   ├── j = 1
   ├── j = 2
   ├── j = 3
   └── j = 4
```

For every iteration of the outer loop, the inner loop completes all of its iterations.

For the given example, how many total iterations are executed if this nested loop?

Biểu đồ / Insight dữ liệu

Dùng khi muốn chuyển từ số liệu sang nhận xét và quyết định

Feature	`while`	`do-while`
Condition check	Before body	After body
Minimum executions	0	1
Type	Pre-test	Post-test

```
do {
    statements;
} while (condition);
```

Main insight

do-while guarantees that the loop body executes at least once.

while
Condition is evaluated first.

do-while
Loop body is executed first.

Multiple-Choice Question

What's the output of the following code?

```
for (int i = 2; i <= 8; i += 2) {
    cout << i << " ";
}
```

A 1 2 3 4 5 6 7 8

B 2 3 4 5 6 7 8

C 2 4 6 8

D 2 4 6 8 10

Correct answer: **B**. The loop starts at 2 and increases i by 2 after every iteration.

Key Takeaways

Kết luận nên ngắn, có thứ tự và dễ nhớ

Foundation	C++ supports three main repetition structures: while, for, and do-while.
Mechanism	Every loop requires a condition that determines whether execution continues.
Application	Counters, sentinel values, accumulators, and nested loops can be used to solve repetitive tasks.
Extension	Use break to terminate a loop and continue to skip the remainder of the current iteration.

A correct loop should always have a clear path toward its termination condition

REFERENCES

Textbook

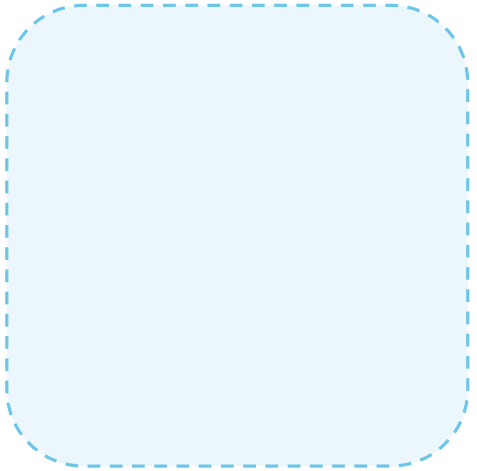
C++ Programming, Ho Chi Minh City University of Technology, 2026
• **Chapter 4 – Repetition Structures in C++**, pp. 43–53

Bài báo / tiêu chuẩn

• [Tên bài báo / chuẩn kỹ thuật]
• [DOI / conference / journal]

Liên kết học tập

• [Link course site / LMS]
• [Link notebook / simulator / dataset]



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Style guide nhanh

Giữ nhận diện Bách Khoa nhất quán trong toàn bộ học phần

Màu chủ đạo



BK Blue
#005BAC

Sky
#6BC6E8

Accent
#F36C21

Navy
#003B73

Light
#F5F8FB

Typography

Tiêu đề slide: Arial 22–28pt, màu BK Blue
Nội dung: Arial 13–17pt, màu xám đậm
Nhấn mạnh: dùng cam, không quá 1–2 điểm / slide

Cách dùng template

1. Duplicate slide layout phù hợp, rồi thay các phần trong [ngoặc vuông].
2. Giữ mỗi slide một thông điệp chính; hạn chế dán nguyên đoạn văn dài.
3. Footer đang để [MÃ MÔN HỌC]; chỉnh trong Slide Master để áp dụng toàn deck.
4. Có thể lưu lại dưới dạng .potx trong PowerPoint: File → Save As → PowerPoint Template.